

TOBIAS LOHSE

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10+ years turning AI capabilities into production systems, architecting pragmatic solutions and leading nimble teams from ML research to shipping great UX.

9/2023 – Present | 2 years 9 months | Remote

Founding Engineer >> Head of Engineering, Orca DB

- Led development of a retrieval-augmented classification platform for tuning training data, adapting online to drift without retraining, and tracing predictions to data.
- Architected low-latency inference at 100 QPS per GKE node with FastAPI and Milvus.
- Implemented high-volume telemetry backend with zero-downtime data migrations.
- Engineered a Postgres job queue for embedding fine-tuning hyperparameter sweeps over custom-built PyTorch losses, delivering major accuracy gains for customers.
- Designed and built a multi-tenant Next.js dashboard with SSO, SCIM, and audit logs.
- Shipped agentic features across a dashboard copilot that acts on the page with Vercel AI SDK, backend agents with Pydantic AI, and a sandboxed dataset-prep coding agent.
- Led 5 engineers, owned product backlog, drove disciplined adoption of agentic coding.
- Streamlined operations across on-call, OTel monitoring, GKE and self-hosted release pipelines, CI harness, automated vulnerability mitigations, and SOC2 compliance.
- Designed the Python SDK and docs to guide developers from quickstart to production.

6/2022 – 7/2023 | 1 year 2 months | Remote

Senior Engineering Manager, Indigo Ag

- Managed team of 7 building tools that paired satellite imagery with soil modeling to target high-potential farmers, driving 80% growth in carbon credit enrollments.
- Coached 2 engineers to promotion, ran book clubs that evolved architecture org-wide, organized 80-person on-site, led full-stack chapter, and improved on-call for 5 teams.

9/2020 – 4/2022 | 1 year 7 months | Remote

Principal Engineer >> Associate Director, Tandem Diabetes Care

- Led project to launch a diabetes management platform for 330k patients worldwide.
- Hired 3 managers and 8 engineers, restructured department with 60 contributors into agile teams, and implemented new performance review system across 10 managers.
- Automated generation of regulatory IEC 62304 filings for SaMD from integration tests.
- Established a unified mobile/web React Native design system and a GraphQL gateway.

10/2016 – 10/2020 | 3 years 11 months | Potsdam, Germany & Remote

Co-founder & CTO, Cinuru Research

- Built white-label React Native apps with gamified loyalty programs for cinemas including the largest German cinema chain, reaching 200k MAUs and a 4.7 rating.
- Led development of Next.js dashboards, GraphQL APIs, and data ingestion workers.
- Implemented newsletter personalization engine that led to 20% ticket sale increases.

1/2016 – 9/2016 | 9 months | Berlin, Germany

Machine Learning Engineer, Aaron AI

- Built NLP call-routing system using TF-IDF, Word2Vec, and random forests.

11/2012 – 1/2015 & 7/2018 – 8/2018 | 2 years 5 months | Berlin, Germany

Web Developer & Designer, Freelance

- Shipped a React Native dating app in one month and built websites for various clients.

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SKILLS

~ 5 YEARS

Machine Learning & AI

Transformers, PyTorch, scikit-learn, NumPy, Pydantic AI, AI SDK, Milvus, RAG, Agent Eval, Fine-tuning, CUDA

~ 9 YEARS

Backend & DevOps

Node.js, Python, REST, GraphQL, RPC, WebSockets, PostgreSQL, SQLite, Redis, OpenTelemetry, Docker, K8s, Terraform, DDD, Event Sourcing, GitHub Actions, GCP, Serverless

~ 13 YEARS

Web & App Development

TypeScript, React Native, Next.js, Tailwind, HTML Emails, Playwright, i18n, a11y, Auth & SSO, Design Systems

EDUCATION

Bachelors supported by Scholarship of Studienstiftung des deutschen Volkes

4/2014 – 12/2015 | FREIE U. BERLIN

Bachelor of Arts in Philosophy & Computer Science

5/2012 – 11/2012 | NORTHWESTERN U.

Bachelor Thesis in Quantum Computing

10/2009 – 5/2012 | U. KONSTANZ

Bachelor of Science in Physics

OPEN SOURCE

OpenAPI HTTPX

CLI for generating typed HTTPX clients from OpenAPI specs

Event Sync

Event sourcing based offline-first state synchronization framework

Styled Native Components

CSS in JS library for React Native